

12 (**long**) Consider the coupled system of differential equations

$$\begin{aligned}\frac{dy}{dt} + 4x &= u(t) \\ \frac{dx}{dt} - y &= w(t)\end{aligned}$$

(a) Show that the Laplace transform $\bar{x}(s)$ of $x(t)$ satisfies

$$\bar{x}(s) = \frac{\bar{u}(s) + s\bar{w}(s)}{s^2 + 4} + \frac{y(0) + sx(0)}{s^2 + 4}$$

where $\bar{u}(s)$ and $\bar{w}(s)$ are the Laplace transforms of $u(t)$ and $w(t)$, respectively. [6]

(b) By using Laplace transforms and not otherwise, determine $x(t)$ and $y(t)$ with initial conditions $x(0) = y(0) = 0$, when $u(t) = 0$ for all t and $w(t)$ is the Heaviside step function,

$$\text{i.e. } w(t) = H(t) = \begin{cases} 1 & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases}. \quad [8]$$

(c) By using Laplace transforms and not otherwise, compute

$$\int_0^t g_1(\tau)g_2(t-\tau) d\tau$$

$$\text{for } g_1(t) = \begin{cases} \cos(2t) & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases} \text{ and } g_2(t) = \begin{cases} t & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases}. \quad [8]$$

(d) Determine $x(t)$ with initial conditions $x(0) = y(0) = 0$, when $u(t) = 0$ for all t and

$$w(t) = \begin{cases} -2t & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases}. \quad [8]$$

END OF PAPER

Part IA 2025

Paper 4: Mathematical Methods

Sections B and C: Numerical Answers

9. (a) (i) $y = x - 2$, line with slope 1 and intercept -2 .

(ii) $x^2 + (y - 4)^2 = 16$, a circle with radius 4 centred at $(0, 4)$

(b) $\int e^{ax} \cos(bx) dx = \frac{e^{ax}}{a^2 + b^2} (a \cos(bx) + b \sin(bx)) + c$

(c) (i) $z = \sqrt{2}e^{i\theta}$, $\theta \in \{\frac{\pi}{12}, \frac{3\pi}{4}, \frac{17\pi}{12}\}$

(ii) $z = \sqrt{2}e^{i\theta}$, $\theta \in \{\frac{\pi}{12}, \frac{7\pi}{12}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{17\pi}{12}, \frac{23\pi}{12}\}$

10. (a) Eigenvalues $1 - a$, a and $1 + a$, eigenvectors $(-1, 1, 0)$, $(0, 0, 1)$ and $(1, 1, 0)$

(b) (i) $\mathbf{A}^n = \frac{1}{2} \begin{bmatrix} \lambda_1^n + \lambda_3^n & -\lambda_1^n + \lambda_3^n & 0 \\ -\lambda_1^n + \lambda_3^n & \lambda_1^n + \lambda_3^n & 0 \\ 0 & 0 & 2\lambda_2^n \end{bmatrix}$

(ii) $e^{\mathbf{A}} = \begin{bmatrix} e \cosh a & e \sinh a & 0 \\ e \sinh a & e \cosh a & 0 \\ 0 & 0 & e^a \end{bmatrix}$

(iii) $e^{\mathbf{A}} \approx \begin{bmatrix} e & ea & 0 \\ ea & e & 0 \\ 0 & 0 & 1 + a \end{bmatrix}$

11. (b) $d = A$

(c) $f(t) = A - A \cos\left(\frac{\pi t}{\alpha T}\right)$

12. (b) $x(t) = \frac{1}{2} \sin(2t)$, $y(t) = \cos(2t) - H(t)$

(c) $\int_0^t g_1(\tau)g_2(t - \tau) d\tau = \frac{1}{4} (H(t) - \cos(2t))$

(d) $x(t) = \frac{1}{2} (\cos(2t) - H(t))$